

Urban Land-Use Efficiency

A Stable Metric for SDG 11.3.1 and Its Economic and Structural Drivers

Sub-national Case Study: Germany (Kreis Level)

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Abstract

This supplement replicates the paper's driver analysis for the Built-up per Capita Rate (BpCR) at the sub-national level, using Germany (district / *Kreis* level) as a case study, and brings in internal (within-country) migration, which the national analysis cannot observe. It asks whether the within-country income-densification and path-dependence results survive at a finer spatial scale.

Keywords: SDG 11.3.1, Built-up per capita rate (BpCR), Sub-national analysis, Germany / Kreise, Internal migration

1. Data and Method

The dependent variable is built exactly as in the main paper, but for German districts (*Kreise*) rather than countries. Urban built-up area V and urban population P come from the GHSL grids (Pesaresi et al., 2024; Schiavina et al., 2023) (urban-masked at GHS-SMOD ≥ 21), aggregated to the GAUL level-2 units for Germany at the five-year epochs 1985–2020; BpCR and urban density are then computed from these totals just as nationally. Two controls are taken from the German regional database INKAR (BBSR, 2024), published by the Federal Institute for Research on Building, Urban Affairs and Spatial Development (BBSR):

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GDP per capita (workplace-based) and the **internal net-migration rate** (*Binnenwanderungssaldo*, per 1,000 inhabitants), the latter being the within-country flow that the national UN measure omits. Urban population share is the GHSL urban population over the INKAR total district population. GAUL districts are matched to INKAR *Kreise* on a cleaned base name, disambiguating same-named *kreisfreie Stadt* / *Landkreis* twins by geographic area, so that 399 of Germany’s 400 districts enter the regressions. The single exception is Freyung-Grafenau, a rural Bavarian-Forest district whose largest settlement reaches only the rural-cluster class (GHS-SMOD 13): no cell attains the urban threshold ($\text{SMOD} \geq 21$), so its urban built-up area and population are zero and BpCR is undefined. This is the same urban definition used throughout the paper, under which a unit with no urban cells is simply not an urban observation, not a matching failure. The panel spans the epochs 1995–2020 for which the INKAR controls are available (2394 district-periods). All specifications carry district and period fixed effects with standard errors clustered by district; the dynamic models add the lagged dependent variable and use Arellano-Bond and Blundell-Bond GMM, as in the main paper.

2. Drivers of Per-Capita Land Use

We estimate the drivers of BpCR at the district level exactly as in the main paper: a two-way fixed-effects model built up control-by-control, then Arellano-Bond (difference) and Blundell-Bond (system) GMM that add the lagged dependent variable to capture path-dependence and correct its Nickell bias. The key addition relative to the national models is **internal (within-country) net migration**, the INKAR *Binnenwanderungssaldo*, which the national panel (international migration only) cannot observe.

Table 1: Dynamic-panel model at the German Kreis level: FE stepwise build-up, Arellano-Bond (difference) and Blundell-Bond (system) GMM.

Variables	Dependent variable: BpCR						
	Fixed Effects					GMM (Bias-Corrected)	
	Base	+ Density	+ Net Migr.	+ GDP	+ Urban %	Arellano-Bond	Blundell-Bond
BpCR[t−1]	+0.2487*** (0.0424)	+0.2266*** (0.0457)	+0.2240*** (0.0445)	+0.2219*** (0.0446)	+0.2191*** (0.0435)	+0.7085*** (0.1221)	+0.8497*** (0.0566)
ln(Urban Density)		-0.0079 (0.0056)	-0.0083 (0.0056)	-0.0091 (0.0058)	-0.0080 (0.0057)	-0.0443*** (0.0054)	-0.0003 (0.0007)
Internal Net Migration			-0.0000* (0.0000)	-0.0000* (0.0000)	-0.0000* (0.0000)	-0.0000 (0.0000)	-0.0000 (0.0000)
ln(GDP per capita)				-0.0019 (0.0016)	-0.0023 (0.0017)	-0.0021 (0.0018)	-0.0004 (0.0005)
Urban Pop. Share					-0.0112 (0.0109)	-0.0263** (0.0107)	-0.0017*** (0.0006)
Observations	2,394	2,394	2,394	2,394	2,394	1,995	2,394
Within R ²	0.065	0.070	0.074	0.075	0.078	–	–
District FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Period FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Instruments	–	–	–	–	–	11	17
Hansen (p)	–	–	–	–	–	0.75	0.00
AR(1) (p)	–	–	–	–	–	0.03	0.00
AR(2) (p)	–	–	–	–	–	0.30	0.01

Note: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Standard errors in parentheses: clustered by district in the fixed-effects columns; two-step GMM with the Windmeijer finite-sample correction (Windmeijer, 2005) in the Arellano-Bond and Blundell-Bond columns. The lagged dependent variable is instrumented by lags 2 to 3 of BpCR, with the instrument matrix collapsed. In the GMM columns the Observations row reports post-transformation equation observations (differenced for Arellano-Bond; the larger stacked level-equation set for Blundell-Bond). Sample: 399 of Germany’s 400 districts; Freyung-Grafenau is excluded because no cell reaches the urban threshold (SMOD of at least 21), leaving its urban built-up and population at zero and BpCR undefined.

Three points emerge. First, per-capita land use is **strongly path-dependent** at the district level too: the lagged dependent variable is large and significant ($\rho \approx 0.71$), echoing the persistence found across countries. Second, the **income-densification** result holds: ln(GDP per capita) is negative and significant across the fixed-effects build-up and in the difference-GMM (Arellano-Bond) estimate, in line with the national finding. Third, **internal net migration** is not significant within districts in this short panel; its clearer association with densification is cross-sectional, with the districts that attract internal migrants being the denser ones.

3. Limitations

Several limitations qualify this case study and mark it as a corroborating local illustration rather than a stand-alone causal analysis.

Coverage and matching. GAUL districts are matched to INKAR by name (399 of Germany’s 400 districts); the only exclusion, Freyung-Grafenau, has no urban-classified cells and hence no defined BpCR. The match relies on a cleaned base name with an area-based rule for same-named city/rural twins rather than an official code, so it is accurate here but would need a code crosswalk to be fully automatic in other settings.

Short, recent panel. The INKAR controls are available only from 1995, so the estimation window is six five-year epochs (1995–2020). This is shorter than the national panel and, together with the five-year spacing, means the models speak to medium-run trajectories rather than short-run adjustment. The brevity matters most for the dynamic estimators: because the GHSL series used here begins in 1985 (BpCR from 1990), the deeper lags that instrument the Arellano-Bond difference equation are available only from 2000 onward, so the GMM is estimated on fewer periods than the within (fixed-effects) columns even though the latter retain all of 1995–2020.

GMM diagnostics. The Arellano-Bond difference estimator is well behaved: the Hansen test does not reject the instruments, first-order serial correlation is present and second-order is absent, as required. The Blundell-Bond system estimator, by contrast, is rejected at this fine and short scale (its Hansen and AR(2) tests fail), so we treat Arellano-Bond as the preferred dynamic estimator and read Blundell-Bond only as a sign-and-significance cross-check.

Internal migration. The internal net-migration control (INKAR *Binnenwanderungssaldo*) is the within-country flow the national analysis cannot observe, but it is not statistically significant within districts in this short panel; its association with densification appears to be cross-sectional (the districts that attract internal migrants are the already-denser ones) rather than a within-district dynamic that six epochs can identify.

Measurement and scope. As in the main paper, the built-up and population layers are satellite-based estimates, the urban area is defined at a fixed Degree-of-Urbanisation threshold (GHS-SMOD of at least 21), and present-day GAUL boundaries are applied throughout; the income and migration controls are administrative aggregates. The estimates are associations, not causal effects, and they describe a single country, so they illustrate rather than establish external validity.

4. Conclusion

This supplement reproduces the paper’s substantive findings at the sub-national (German district) scale, across 2394 district-periods. Per-capita urban land use is strongly path-dependent within districts (Arellano-Bond $\rho \approx 0.71$), echoing the persistence found across countries, and it declines with income: getting richer is associated with less built-up land per resident, the same compact-city, income-densification relationship the national panel shows. The result is therefore not an artefact of country-level aggregation; it holds at a much finer spatial resolution and within a single national institutional setting.

The case study also adds the internal migration that the national analysis cannot see. The within-country net-migration flow is not statistically significant in this short, six-epoch panel; its sign and the broader cross-sectional pattern are consistent with in-migrants being absorbed into the existing urban stock rather than driving outward expansion, but identifying that channel would require a longer panel. Taken together, the German evidence corroborates, at a scale the national panel cannot reach, that economic growth within a place is accompanied by densification rather than sprawl.

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